

AICTE Training and Learning (ATAL)

Online Faculty Development Program (FDP) on
High performance Analog and Mixed signal VLSI System design

Organized by Indian Institute of Technology Dharwad (IIT Dharwad)



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भारतीय प्रौद्योगिकी संस्थान धारवाड

Indian Institute of Technology Dharwad

13th to 17th of December, 2021

Sponsor: *All India Council for Technical Education (AICTE)*

FDP Coordinator: Dr. Naveen Kadayinti

Assistant Professor, Dept. of Electrical Engineering,

Indian Institute of Technology Dharwad

Registration Link: <https://www.aicte-india.org/atal>

About IIT Dharwad

Indian Institute of Technology Dharwad (IIT Dharwad) is an autonomous premier Science and Technology Institute established in 2016. IIT Dharwad offers Btech and MS programmes in Electrical engineering, Computer science engineering and Mechanical Engineering. IIT Dharwad has established research facilities and labs in Physics, Chemistry, Biological sciences, Mathematics and Humanities apart from Engineering departments and offers PhD programmes in all departments.

Department of Electrical Engineering

Electrical Engineering (EE) at IIT Dharwad focuses mainly on fundamentals of building physical devices and exploring the mathematical sides of EE. With faculty from varying research backgrounds, the department encourages multi-disciplinary research allied with computer science, applied physics, biotechnology, chemistry, and computation. Undergraduate and graduate students will study and work on some of the exciting research problems in a wide variety of subjects, both theoretical and experimental. The EE Department has active collaborations with numerous industries and academic institutions, both in India and abroad. The EE Department highly encourages graduate/undergraduate students to take up internships in research labs across the globe. The faculty at EE Department are mainly working in the areas of next generation wireless communication such as IoT, 5G and beyond, communication networks, performance modeling, resource allocation, sensors, circuits and devices, control engineering, machine learning and deep neural networks, signal processing, speech processing, power systems and power electronics. These research activities are carried out in our advanced laboratory facilities at the institute.

Objectives of the FDP

The main objective of this course is to develop a good understanding of analog and mixed signal integrated circuit design and the unique challenges involved in such designs. The course will discuss problems associated with mixed signal circuits and discuss general principles for high performance design considering circuits, packaging as well as PCB design choices.

Content of the Faculty Development Program (FDP) on “High performance Analog and Mixed signal VLSI System design”

1. CMOS amplifiers and opamp design – architectures for high bandwidth, low offset, sample and hold circuits, comparators design
2. Power supply design (on-chip and on board) for mixed signal circuits and systems
3. Charge pump circuits
4. Voltage reference circuits
5. High speed circuits and layout design
6. Clock data recovery circuits and phase locked loops
7. Testing of high performance circuits

Note: Lab sessions using open source tools will be conducted for reinforcing the circuits discussed in the theory sessions.

Tentative Program Schedule for FDP

Time Date	0930 to 1100	Break: 1100 to 1115	1115 to 1245	Break: 1245 to 1400	1400 to 1530	Break: 1245 to 1400	1545 to 1715
13/12/2021	Session 1A: Introduction to the course, Broad picture of Mixed signal circuits		Session 1B: CMOS amplifiers design, 2 stage Opamps,		Session 2: Miller and feed-forward compensation		Session 3: Lab session on amplifier and opamp design
14/12/2021	Session 4A: Fully differential amplifiers		Session 4B: Design of Voltage reference circuits		Session 5: Sample and hold circuits		Session 6: Lab session on Voltage reference design
15/12/2021	Session 7A: Switched capacitor circuits		Session 7B: High speed Comparators design		Session 8: Power supply design for Mixed signal systems		Session 9: Lab session on Sample and hold circuits
16/12/2021	Session 10A: High speed clock and data recovery circuits		Session 10B: CML circuits design		Session 11: High speed circuits and layout design		Session 12: Lab session on Layout design
17/12/2021	Session 13A: Testing of mixed signal integrated circuits		Session 13B: Invited talk by Industry expert on Wide Tuning Range CMOS VCOs		Session 14: Values and ethics session by expert.		Session 15: Online test for participants followed by Concluding session.

Resource Persons

1. Dr. Vinayak Hande, Senior Researcher at Integrated systems and Circuit Design, Carinthia University of Applied Science (formerly at IIT Ropar)
2. Dr. Mahendra Sakare, IIT Ropar
3. Dr. Naveen Kadayinti, IIT Dharwad
4. Dr. Siva Rama Krishna Rudrubati, Staff Engineer at Mediatek Wireless Limited, United Kingdom.
5. Dr. Aditya Pandurangi

Target Participants

1. Faculty Members in Electronics and Communications and related areas from AICTE Accredited Institutions. PG Scholars in these disciplines.
2. Members in technical positions related to circuits design from Industry and Govt organizations like ISRO, BEL etc.

General Information

1. There is no registration fee for participants.
2. Minimum 80% attendance and 60% marks in the test at the end of the FDP are compulsory for certification.

Course Coordinator:

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